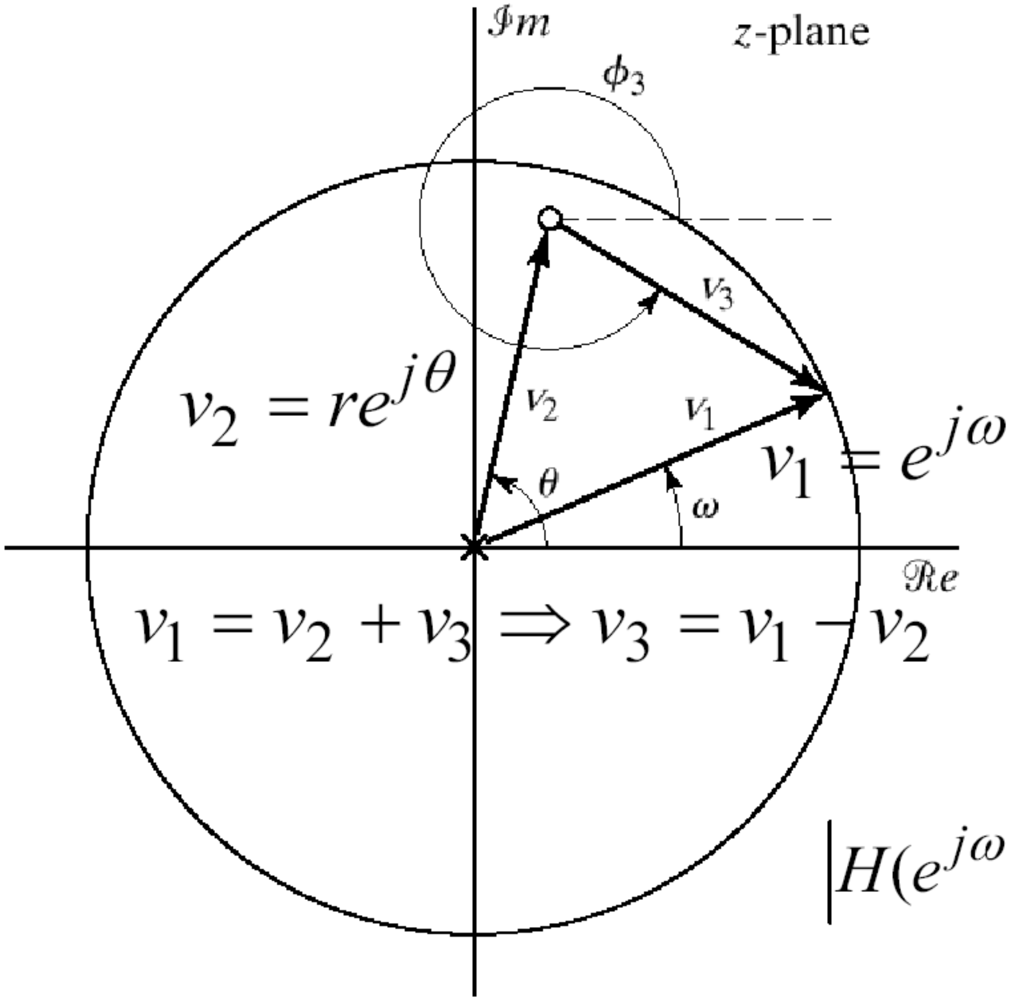
Digital Signal Processing

Lecture 7

Pole Zero plot of LTI System

Dr. Tahir Zaidi

Pole Zero Plot



$$H(z) = (1 - re^{j\theta}z^{-1})$$

$$= \frac{z - re^{j\theta}}{z}$$

$$H(e^{j\omega}) = \frac{e^{j\omega} - re^{j\theta}}{e^{j\omega}}$$

$$= \frac{v_3}{v_1}$$

$$\left|H(e^{j\omega})\right| = \frac{\left|e^{j\omega} - re^{j\theta}\right|}{\left|e^{j\omega}\right|} = \frac{|v_3|}{|v_1|} = |v_3|$$

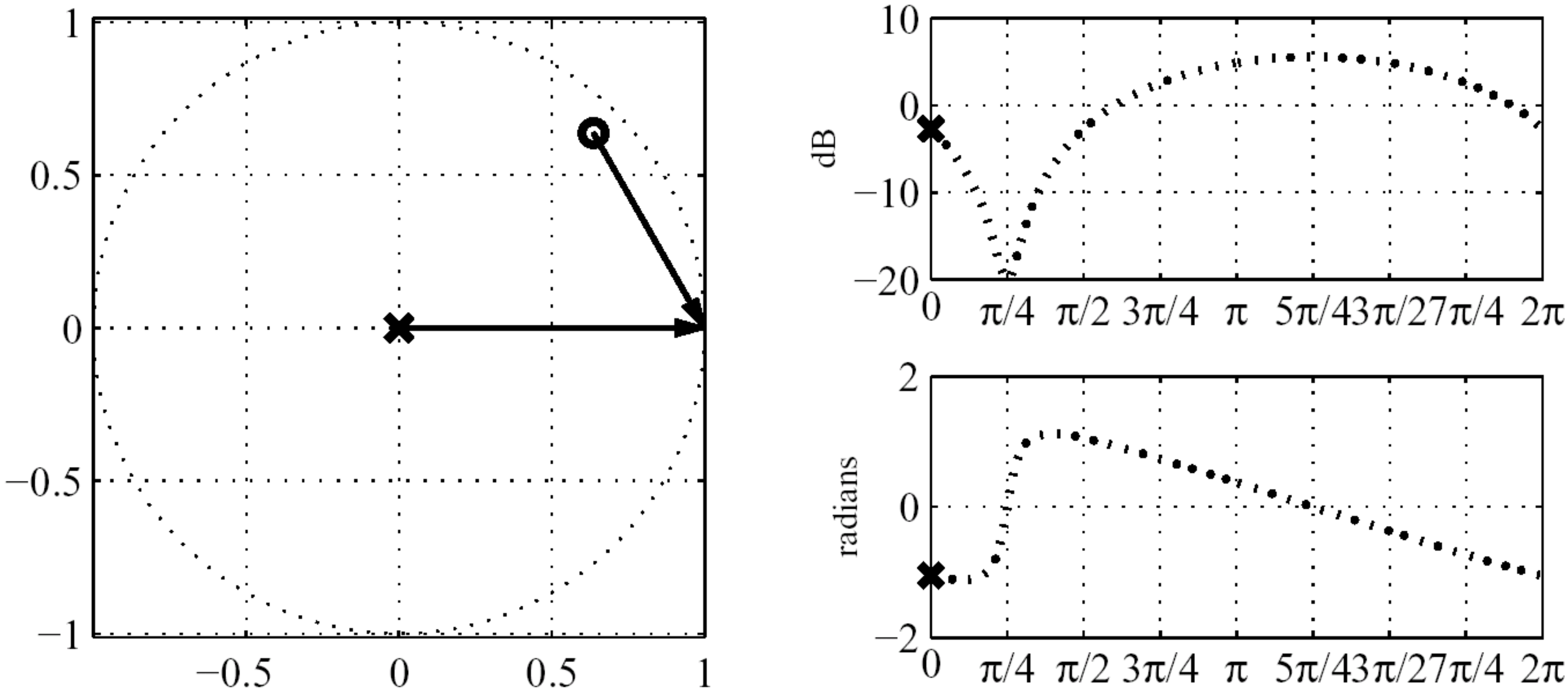
Frequency response of one zero system

$$\left|H(e^{j\omega})\right| = \left|1 - re^{j\theta}e^{j\omega}\right| = \frac{\left|e^{j\omega} - re^{j\theta}\right|}{\left|e^{j\omega}\right|}$$

- Poles Vectors and Zero Vectors
- Magnitude: ratio of zero-vector magnitudes to pole-vector magnitude
- Phase: difference of zero-vector phases to pole-vector phases

Frequency Response Evaluation

One zero at $.9e^{\pi/4}$



Frequency Response Evaluation

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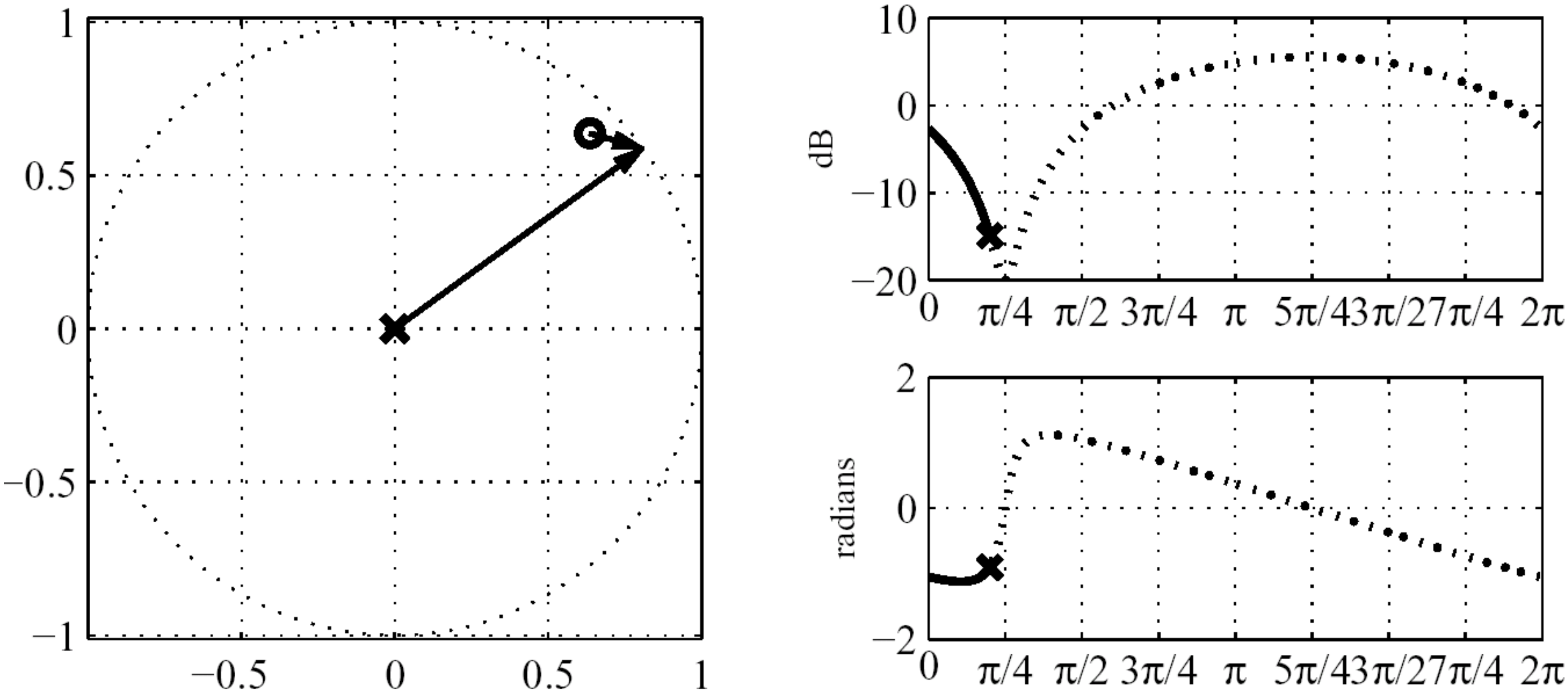

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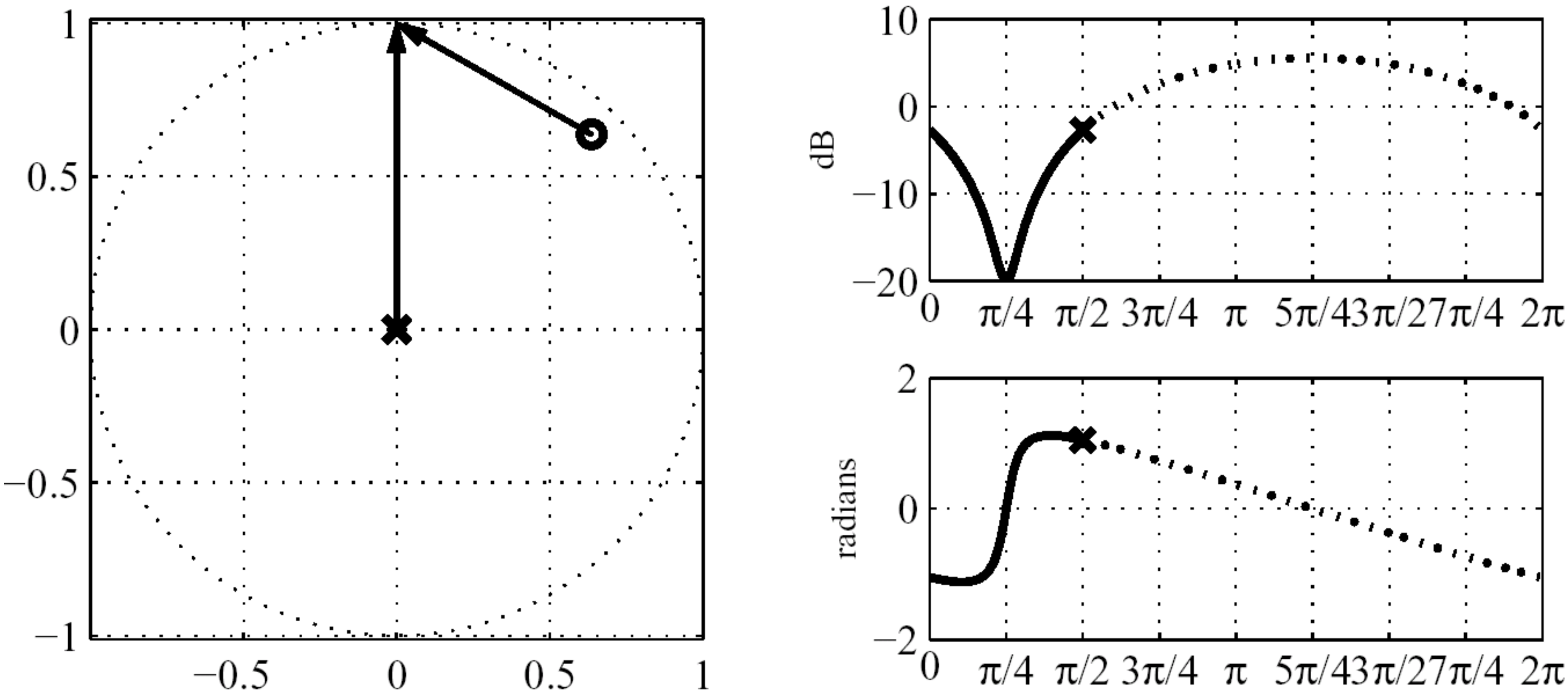
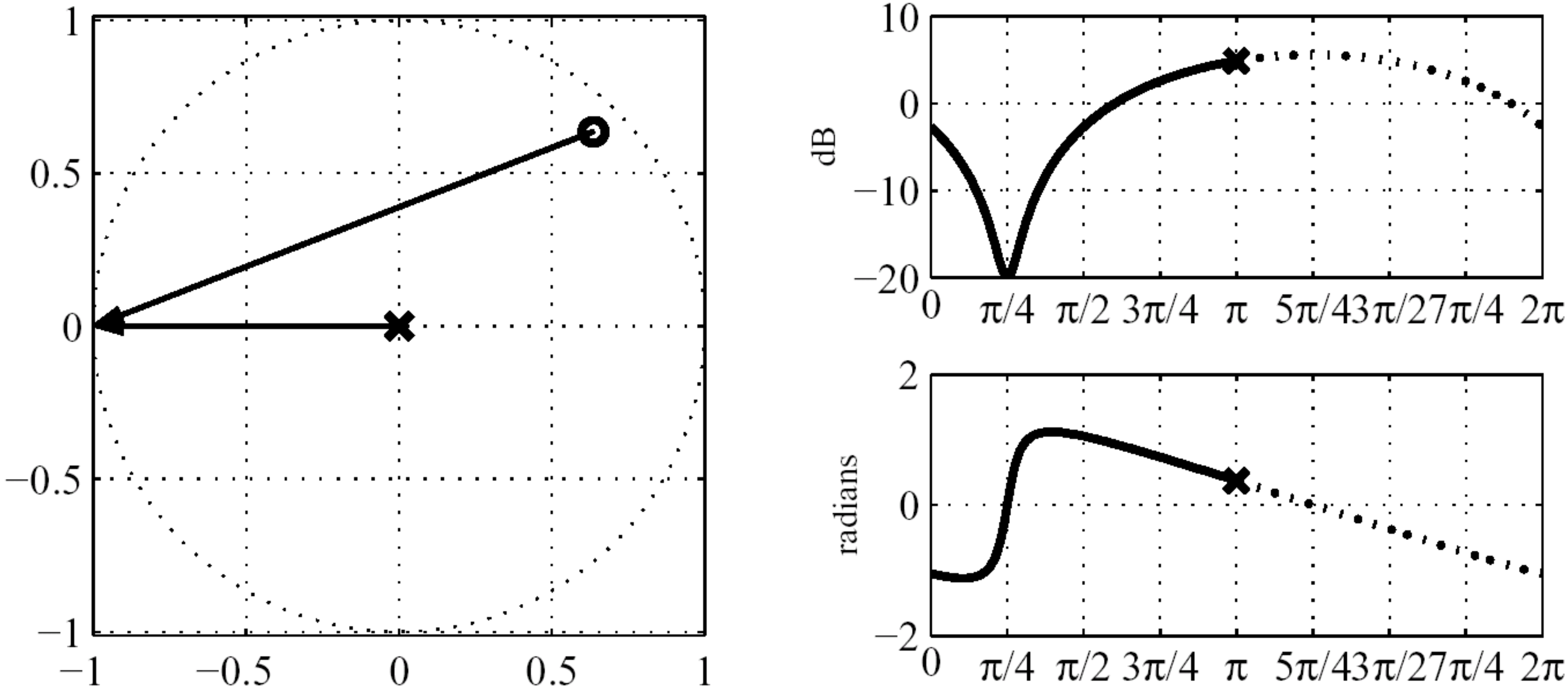
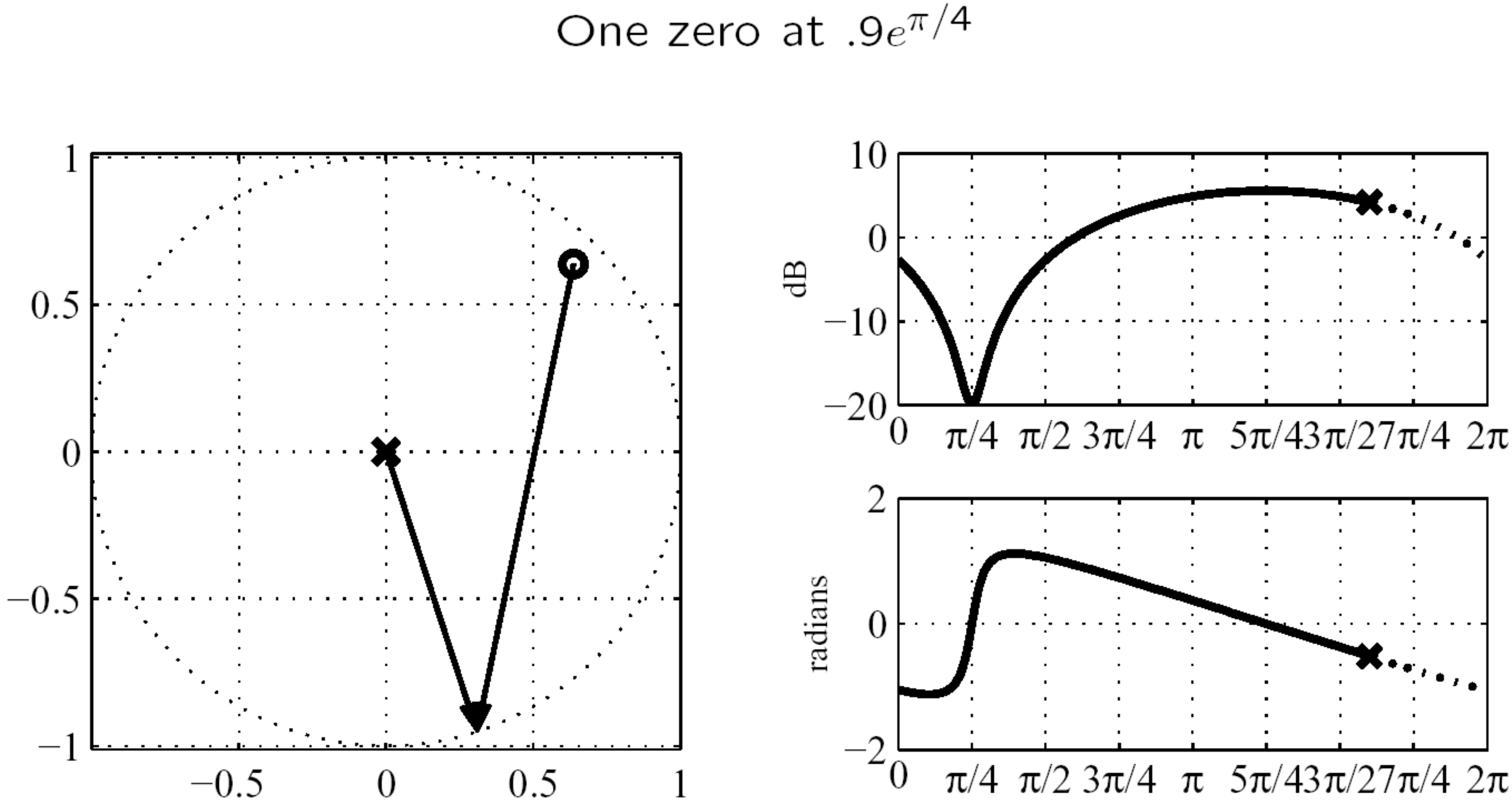


Frequency Response Evaluation

One zero at $.9e^{\pi/4}$



Frequency Response Evaluation



FIR vs IIR

$$H(z) = \sum_{r=0}^{M-N} B_r z^{-r} + \sum_{k=1}^{N} \frac{A_k}{1 - d_k z^{-1}}$$

$$h[n] = \sum_{r=0}^{M-N} B_r \delta[n-r] + \sum_{k=1}^{N} A_k (d_k)^n u[n]$$

FIR part

IIR part

Frequency Response of Rational System Functions

Magnitude: $$|H(e^{j\omega})| = \left|\frac{b_0}{a_0}\right| \cdot \frac{\prod_{k=1}^{M} |(1-c_k e^{-j\omega})|}{\prod_{k=1}^{N} |(1-d_k e^{-j\omega})|}$$

Phase: $$\angle H(e^{j\omega}) = \angle\frac{b_0}{a_0} + \sum^{M} \angle(1-c_k e^{-j\omega}) - \sum^{N} \angle(1-d_k e^{-j\omega})$$

Delay: $$\text{grd}\, H(e^{j\omega}) = -\sum^{M} \frac{d}{d\omega} \arg\angle(1-c_k e^{-j\omega}) + \sum^{N} \frac{d}{d\omega} \arg\angle(1-d_k e^{-j\omega})$$

(note) arg : continuous phase

ARG : its principal value in $(-\pi, \pi)$

$$\arg H(e^{j\omega}) = \text{ARG}\, H(e^{j\omega}) + 2\pi r(\omega)$$

Frequency Response Function

$$H(e^{j\omega}) = \left|H(e^{j\omega})\right| e^{j\angle H(e^{j\omega})}$$

- Log-magnitude (in dB)

$$20\log_{10}\left|H(e^{j\omega})\right|$$

- Phase (in radians)

$$\angle H(e^{j\omega}) = \arg\left[H(e^{j\omega})\right]$$

- Group delay (in samples)

$$\tau(\omega) = \mathrm{grd}\left[H(e^{j\omega})\right] = -\frac{d}{d\omega}\left\{\angle H(e^{j\omega})\right\}$$

Frequency Response

$$x[n] \longrightarrow \boxed{H(e^{j\omega})} \longrightarrow y[n]$$

$$Y(e^{j\omega}) = H(e^{j\omega})X(e^{j\omega})$$

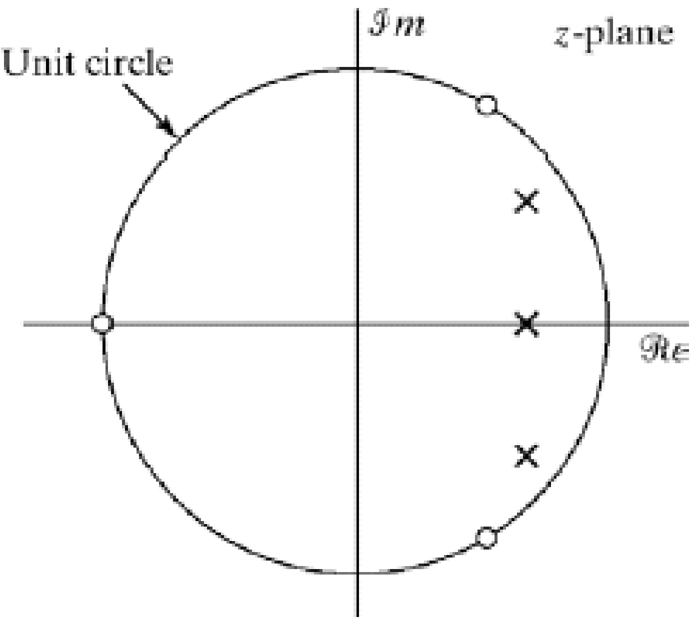
- Log-magnitude response (dB) / Gain:

$$20\log_{10}|Y(e^{j\omega})| = 20\log_{10}|X(e^{j\omega})| + 20\log_{10}|H(e^{j\omega})|$$

- phase response/shift:

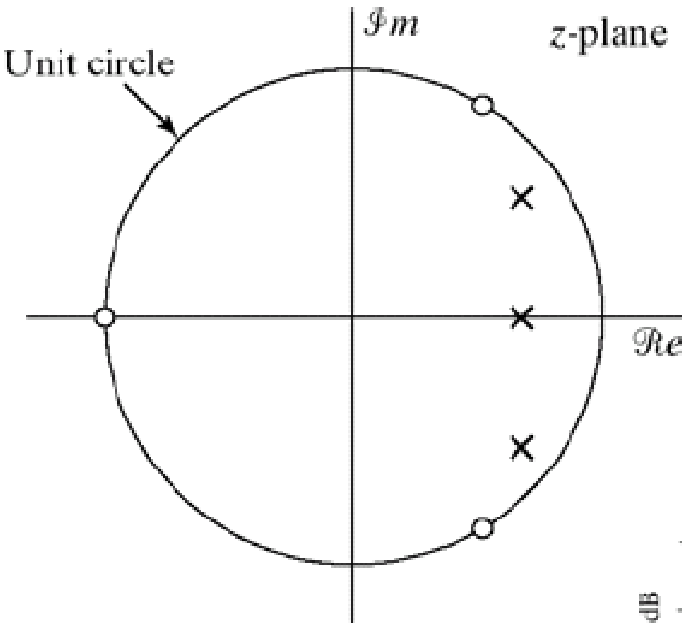
$$\angle Y(e^{j\omega}) = \angle X(e^{j\omega}) + \angle H(e^{j\omega})$$

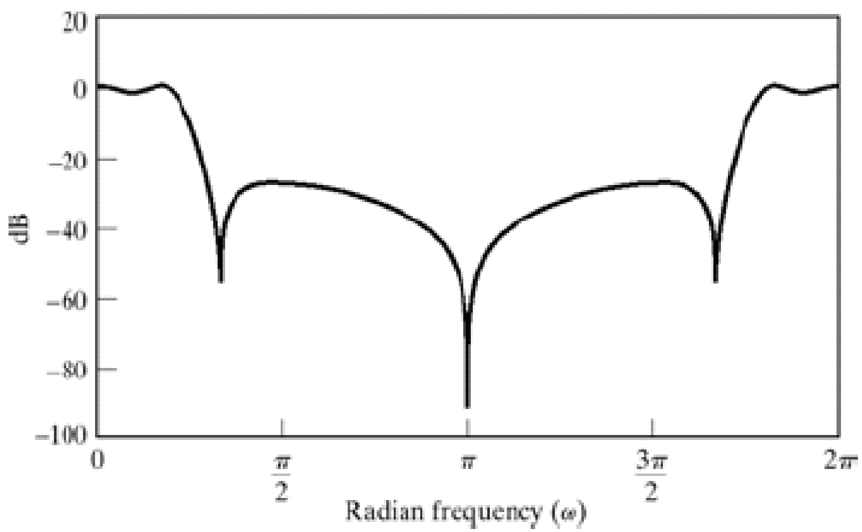
Example of an IIR Filter



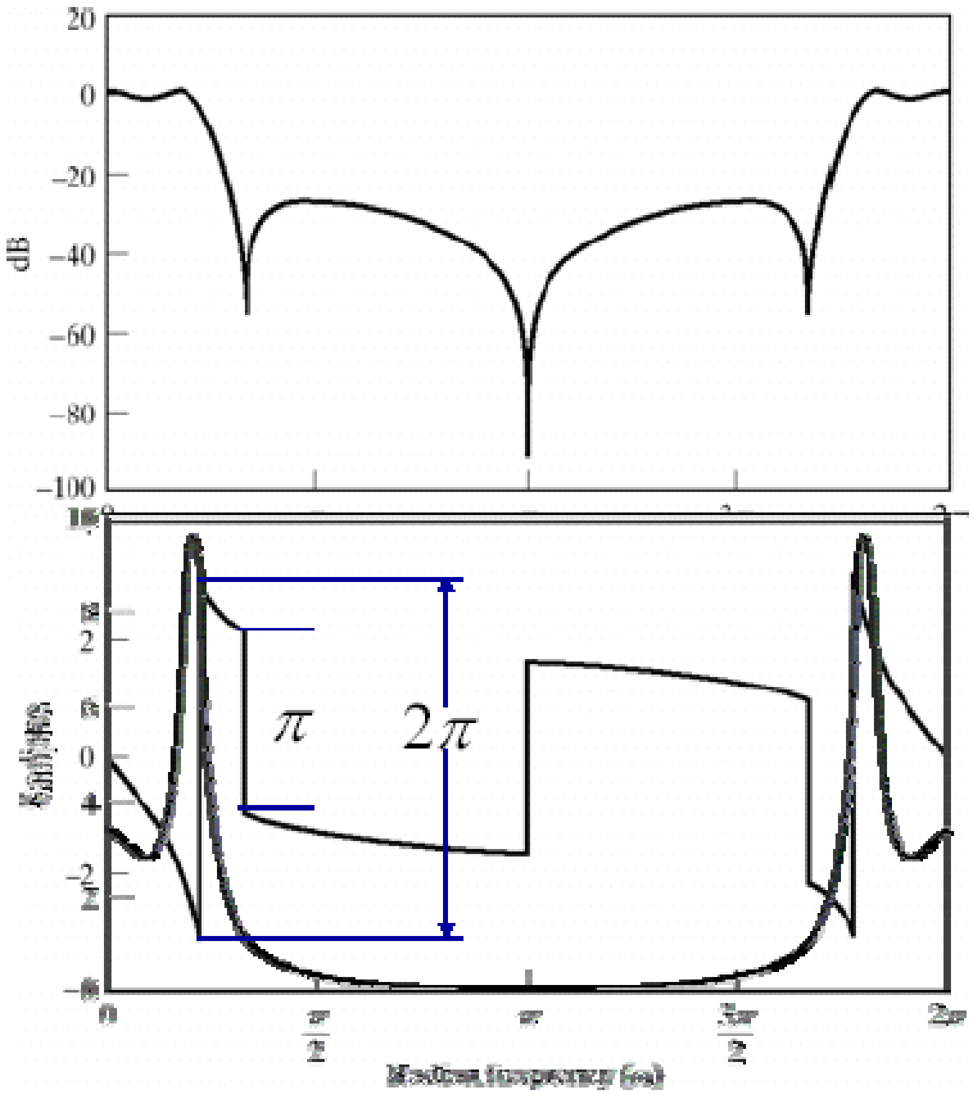
$$H(z)=\frac{0.05634(1+z^{-1})(1-1.0166z^{-1}+z^{-2})}{(1-0.683z^{-1})(1-1.4461z^{-1}+0.7957z^{-2})}$$

Frequency Response

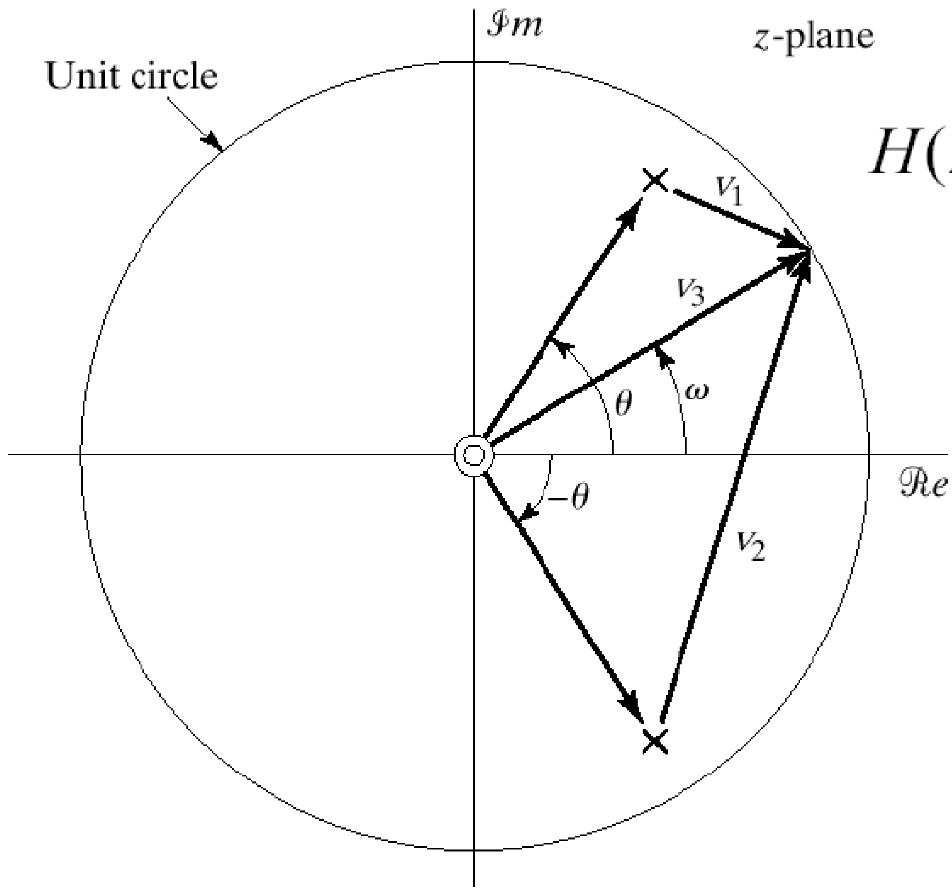




Frequency Response



Example

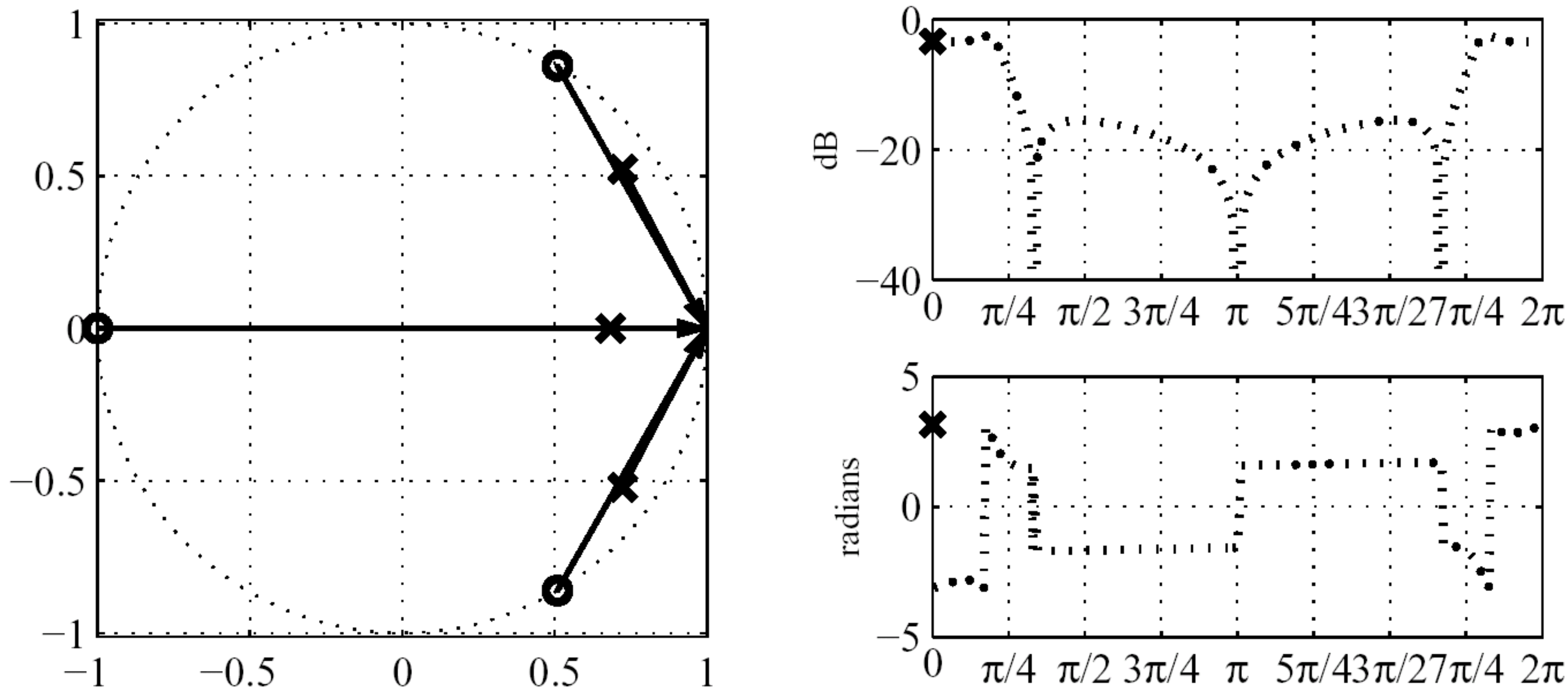


$$H(z) = \frac{z^2}{\left(z - re^{j\theta}\right)\left(z - re^{-j\theta}\right)}$$

$$\left|H(e^{j\omega})\right| = \frac{\left|v_3\right|^2}{\left|v_1\right| \cdot \left|v_2\right|}$$

Frequency Response Evaluation

Third order system with 3 poles and 3 zeros



Frequency Response Evaluation

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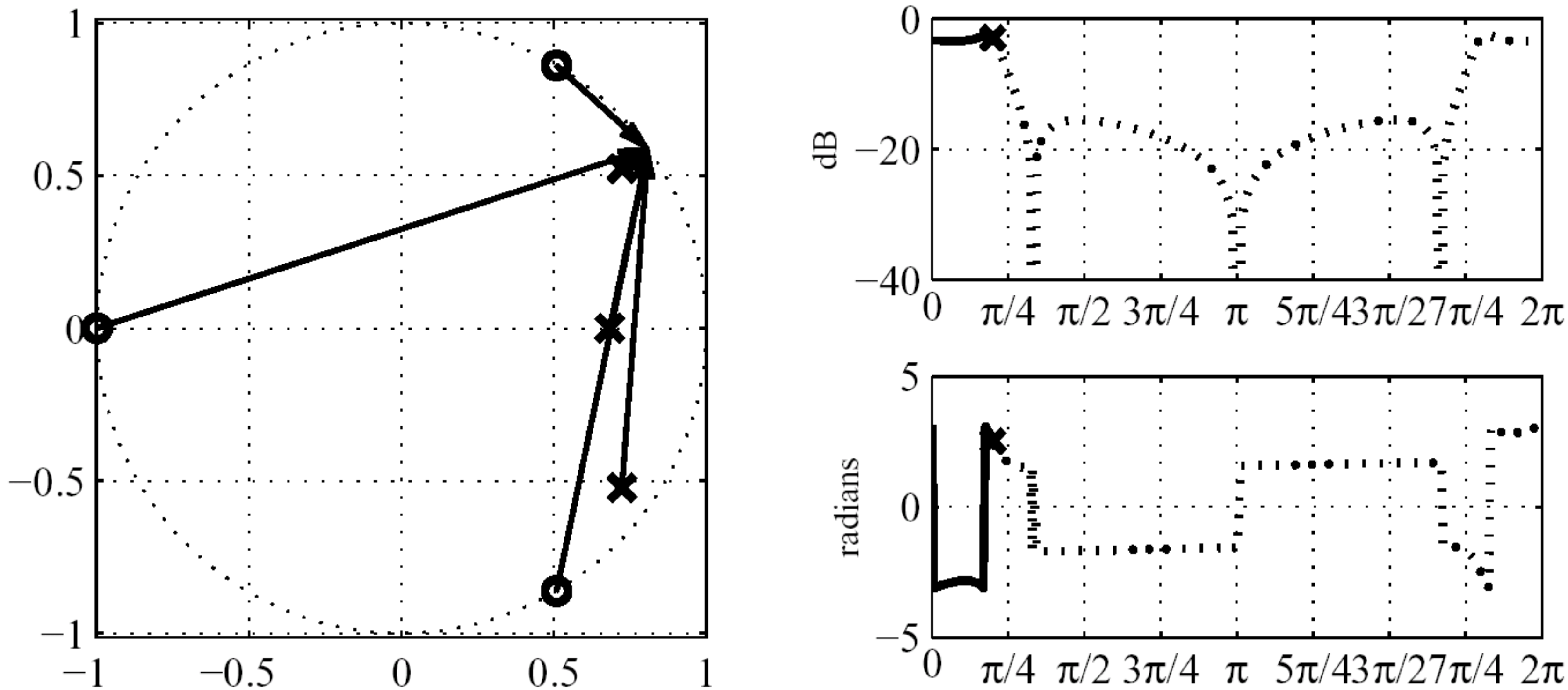

Frequency Response Evaluation

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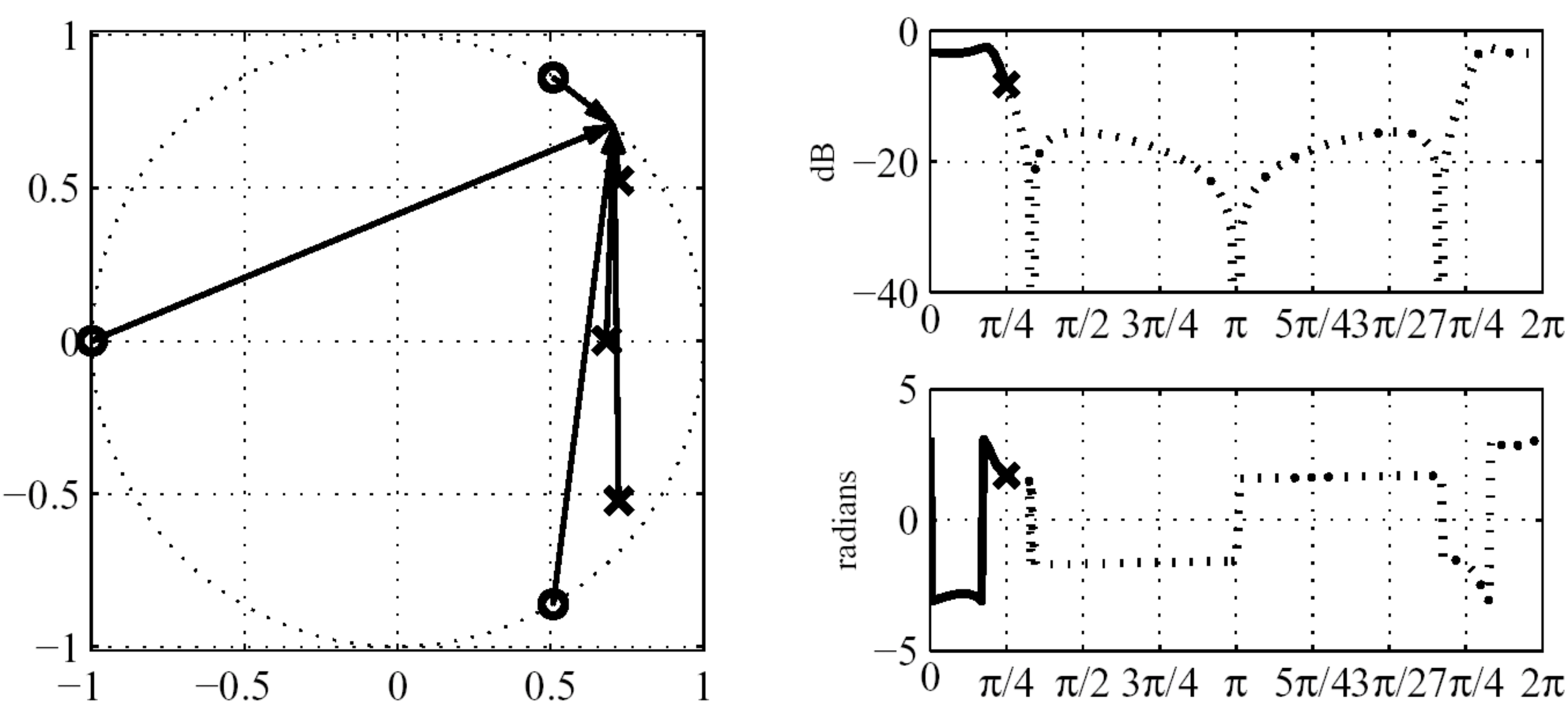

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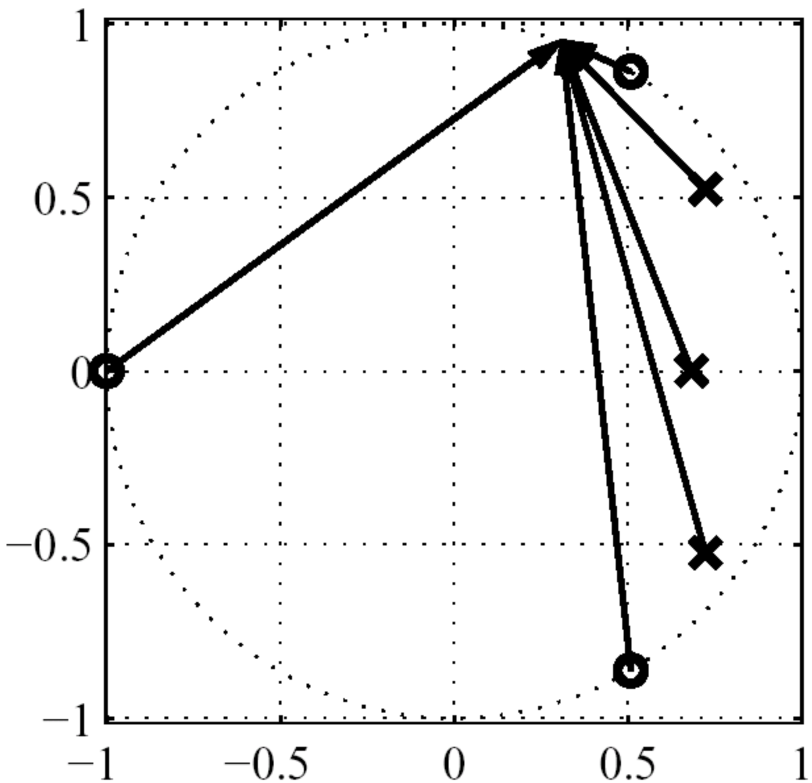
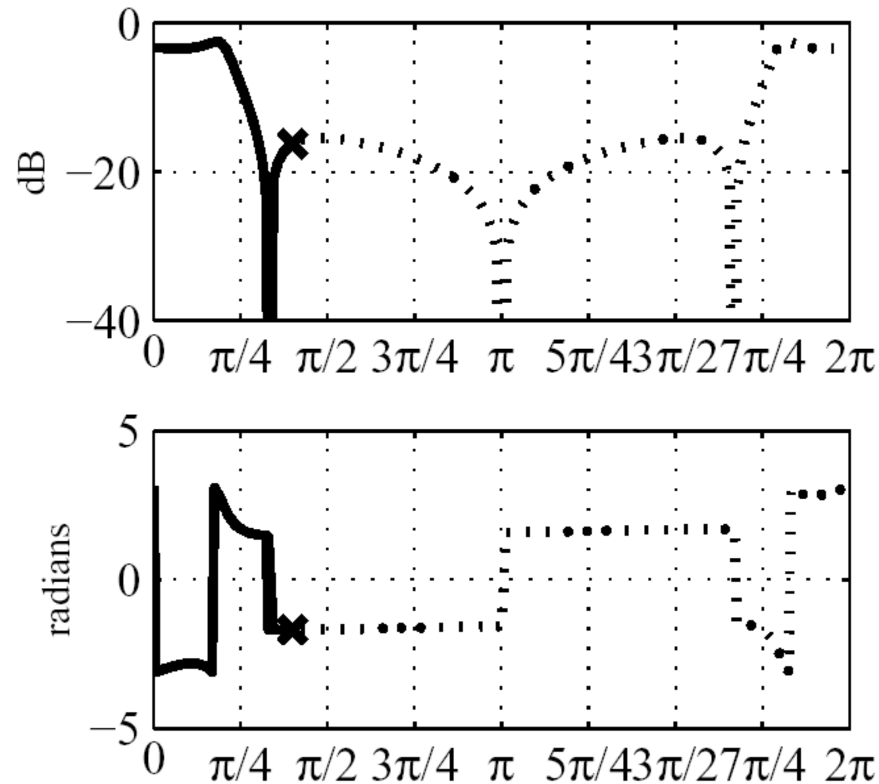
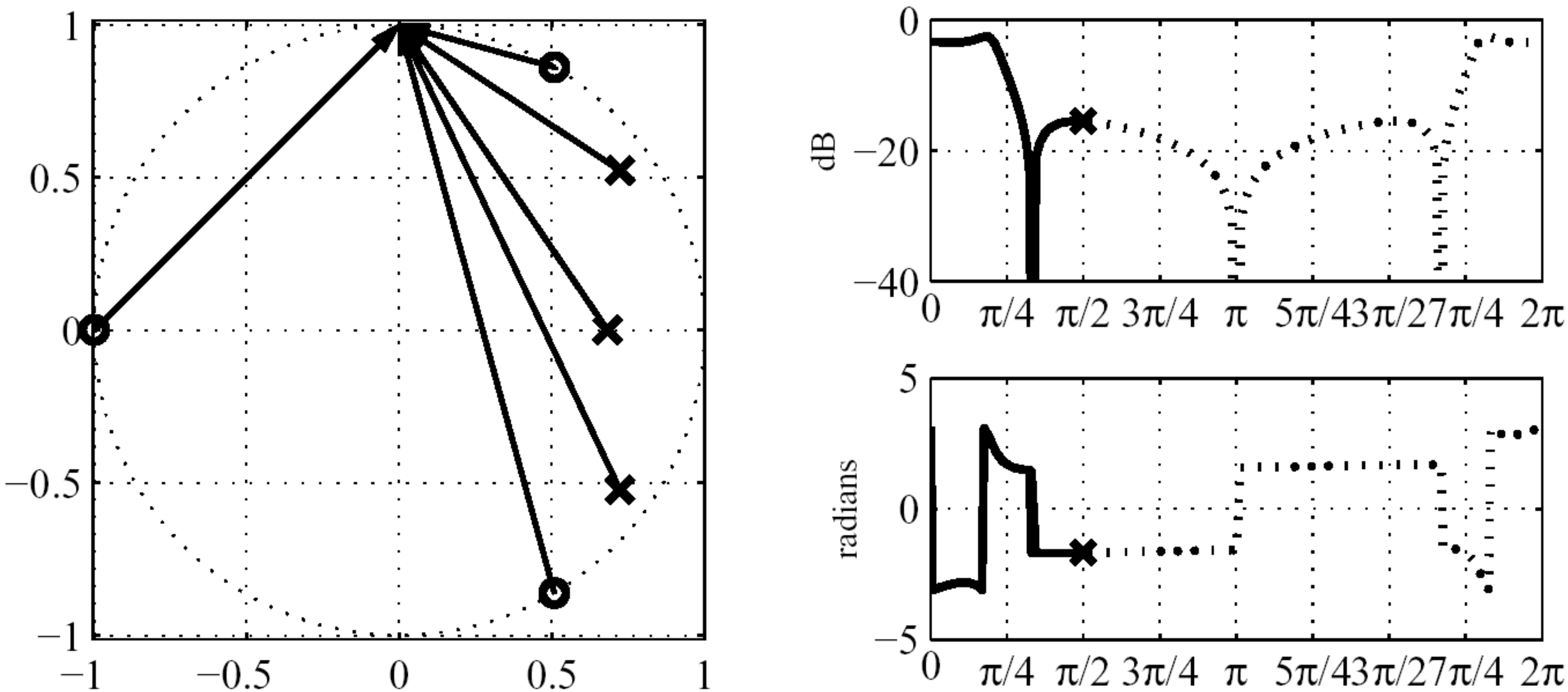




Frequency Response Evaluation

Third order system with 3 poles and 3 zeros



All-pass Systems

- System with unit magnitude response
- Basic building Block:

$$A(z) = \frac{z^{-1} - a^*}{1 - az^{-1}}$$

Pole in a , zero in $1/a^*$

- Real co-efficient all-pass:

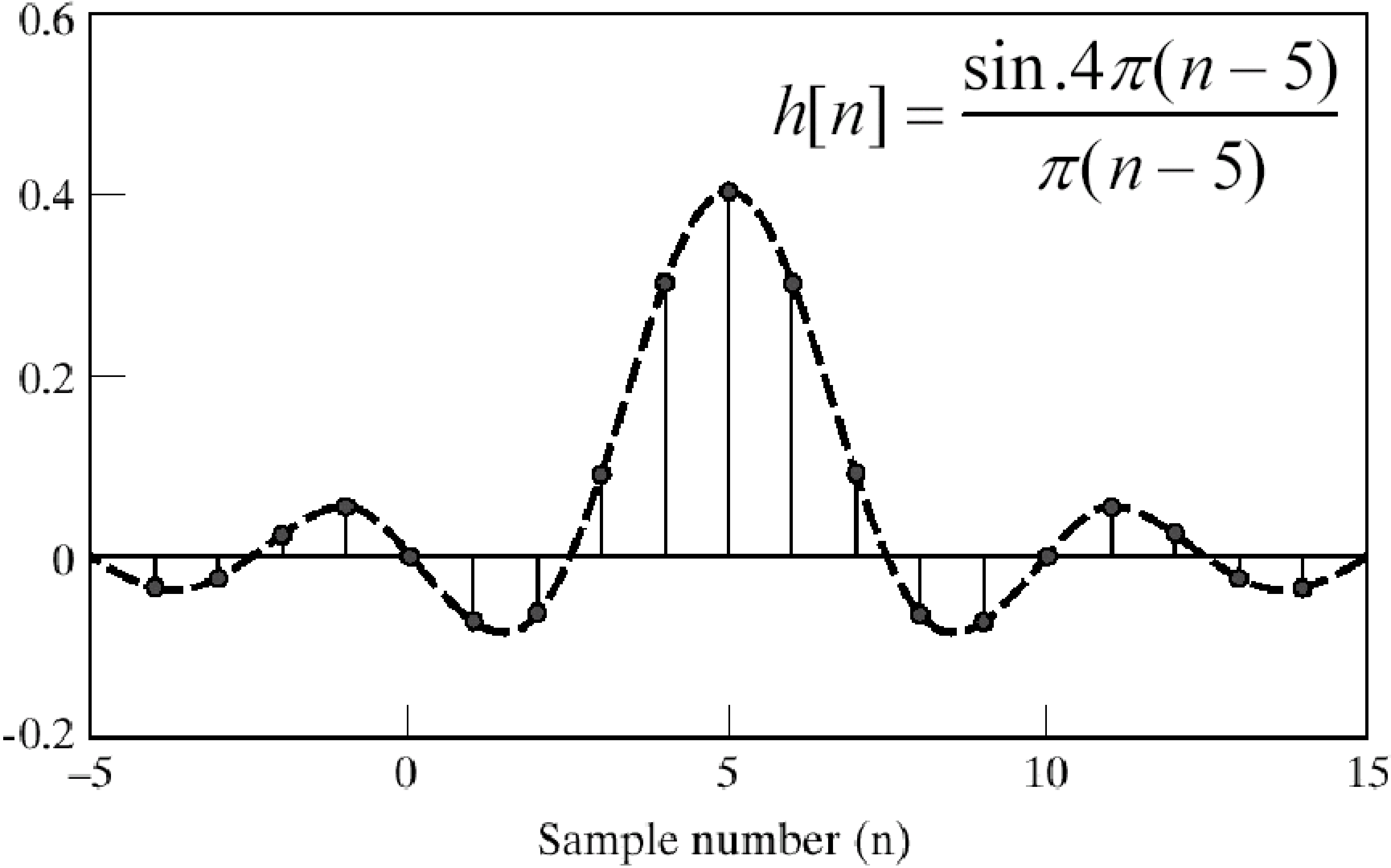
$$A(z) = \prod_k \frac{z^{-1} - a_k^*}{1 - a_k z^{-1}} \prod_k \frac{(z^{-1} - b_k^*)(z^{-1} - b_k)}{(1 - b_k z^{-1})(1 - b_k^* z^{-1})}$$

Basic LP system

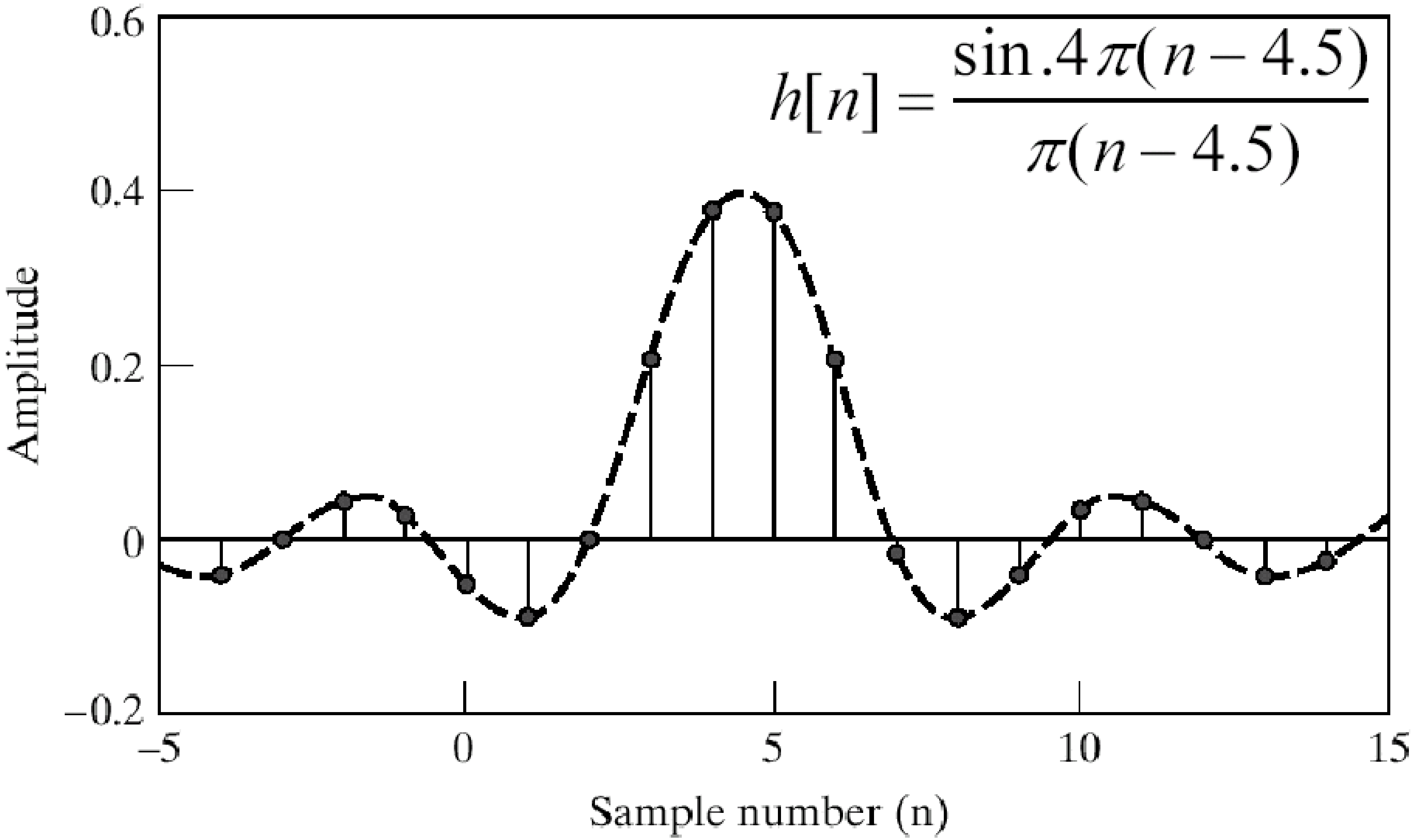
$$\left|H(e^{j\omega})\right|e^{-j\omega\alpha}$$

- Constant group delay α
- Impulse response symmetric about 2α if 2α integer
- No symmetry if $2\alpha \notin \mathrm{Z}$

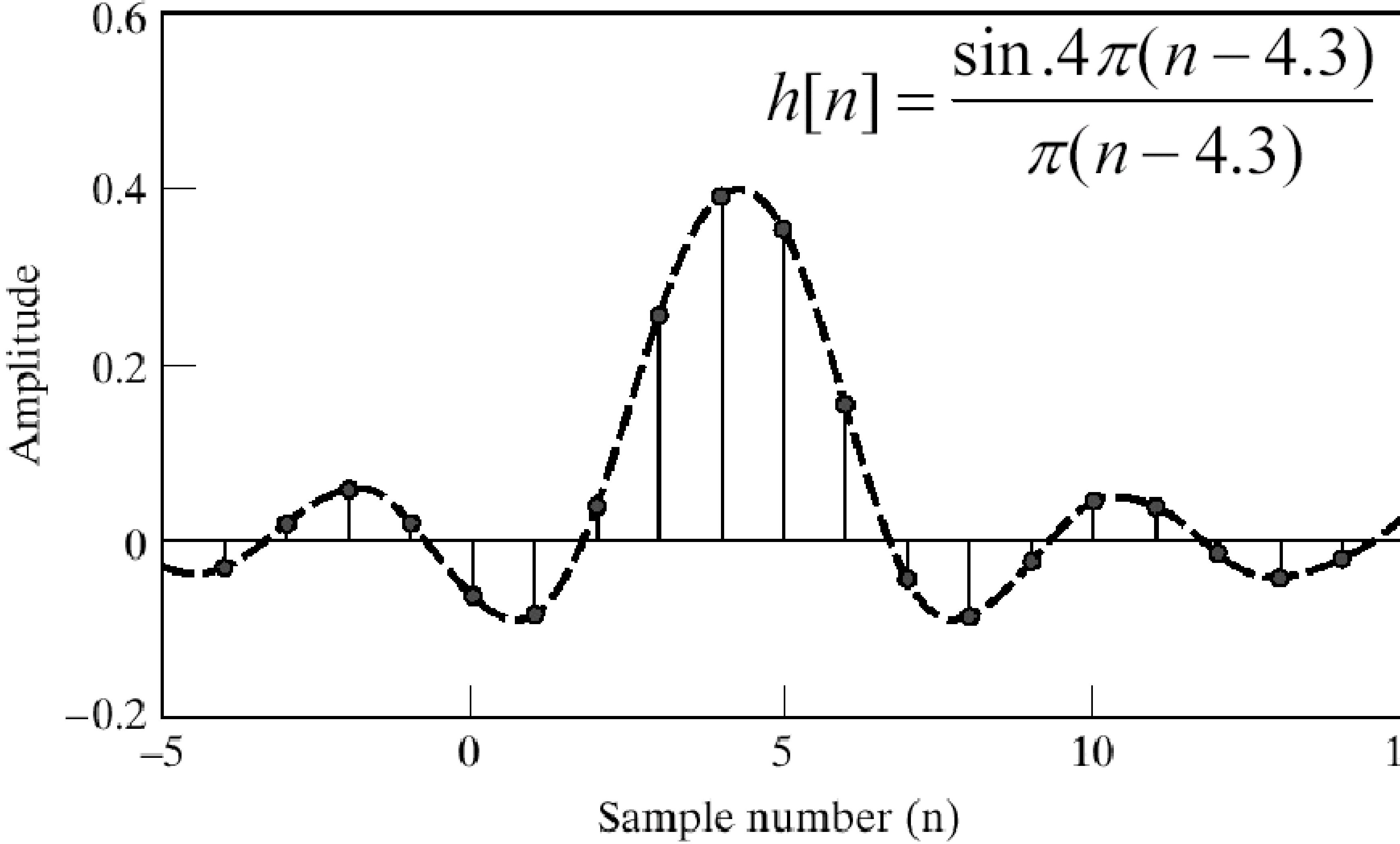
Ideal low-pass with Delay



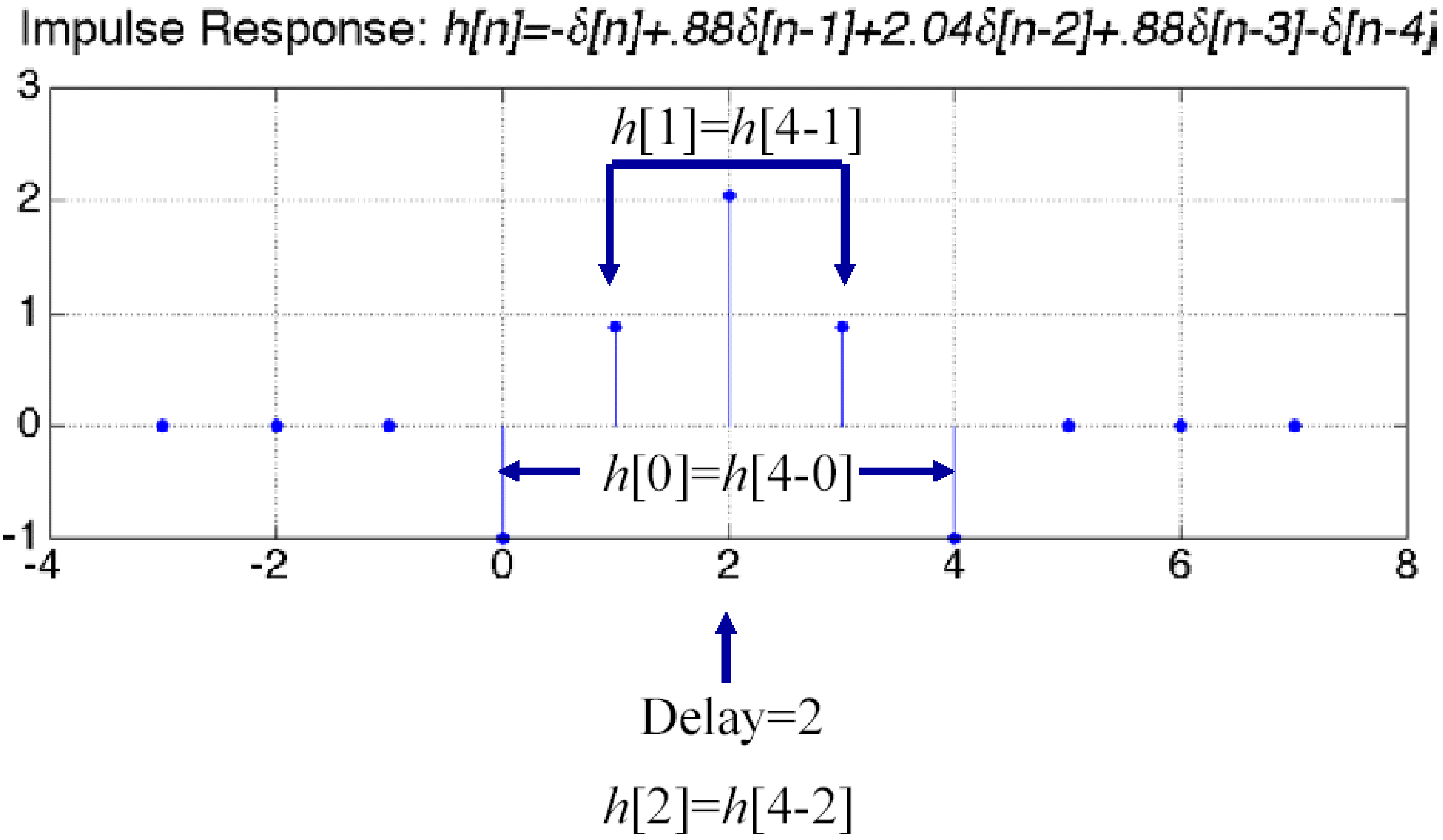
Ideal lowpass with Linear Phase



Ideal Lowpass with Linear Phase



Impulse Response



Pole and Zero Locations

$$H(z) = -1 + .88z^{-1} + 2.04z^{-2} + .88z^{-3} - z^{-4}$$

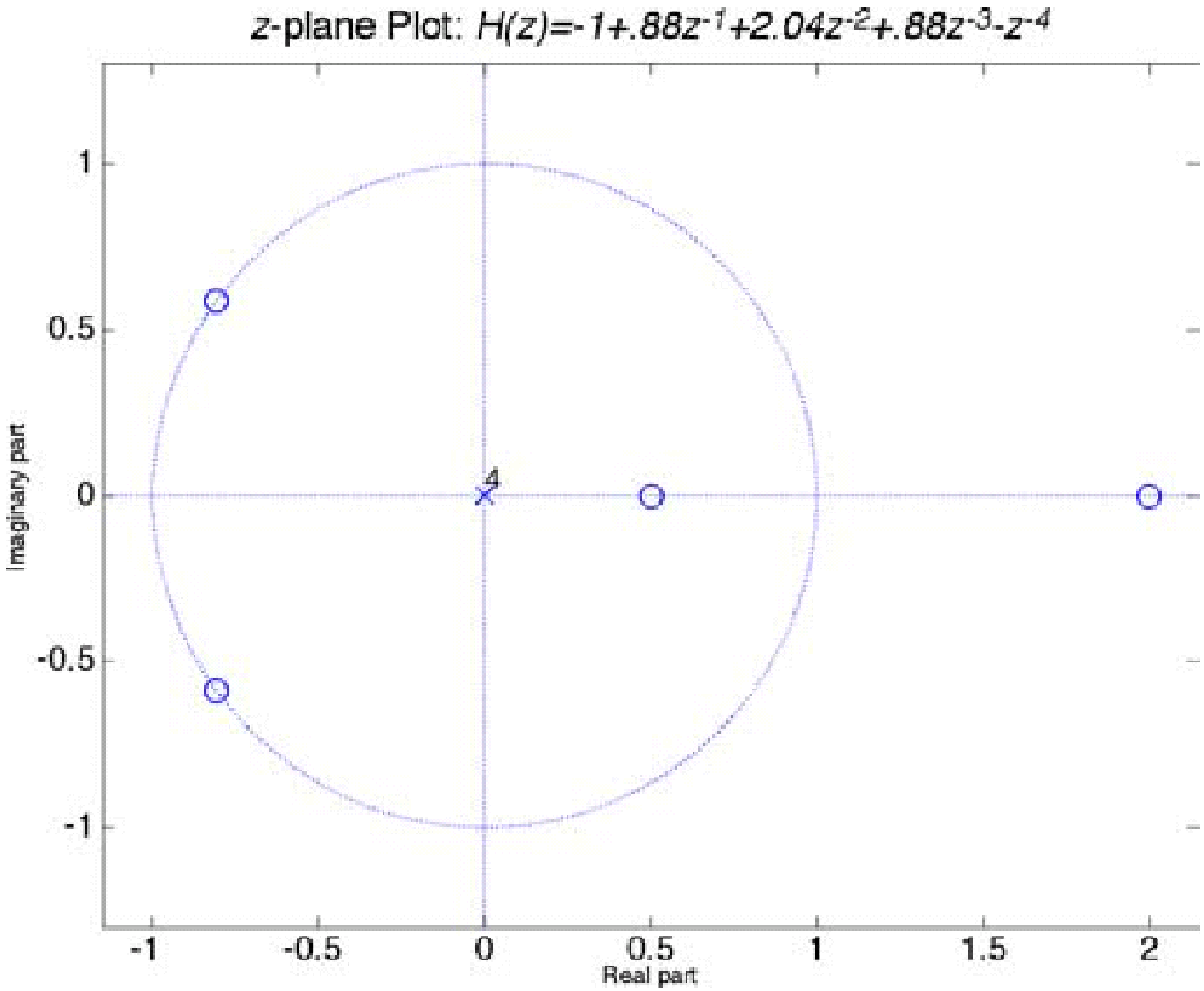
$$H(z) = \frac{-z^4 + .88z^3 + 2.04z^2 + .88z - 1}{z^4}$$

$$H(z^{-1}) = -z^4 + .88z^3 + 2.04z^2 + .88z - 1$$

$$H(z^{-1}) = z^4 H(z)$$

$$H(z_0) = 0 \quad \Leftrightarrow \quad H(z_0^{-1}) = 0$$

Pole-Zero Plot



Frequency Response

$$H(e^{j\omega}) = -1 + .88e^{-j\omega} + 2.04e^{-j\omega 2} + .88e^{-j\omega 3} - e^{-j\omega 4}$$

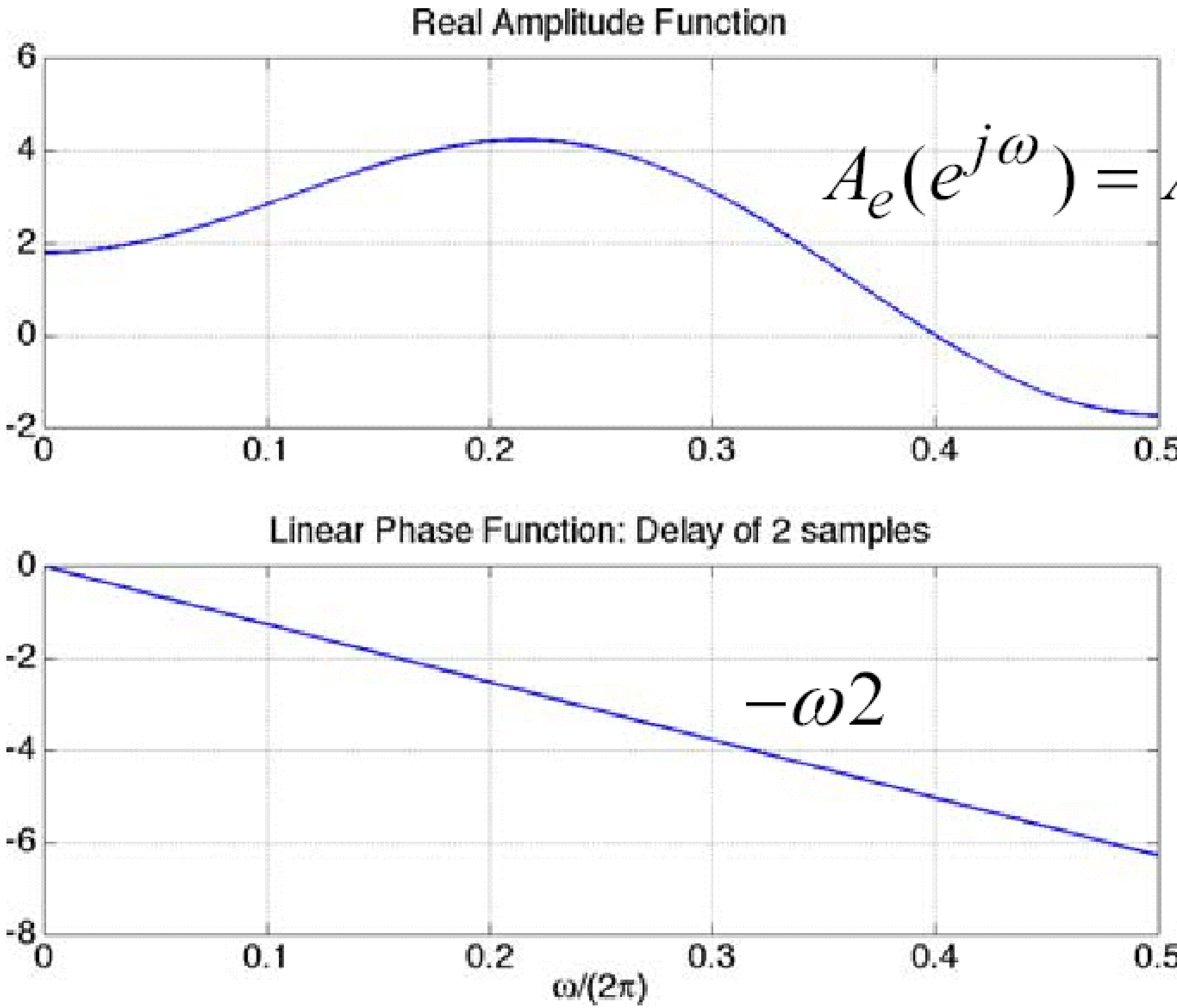
$$H(e^{j\omega}) = \left(-e^{j\omega 2} + .88e^{j\omega} + 2.04 + .88e^{-j\omega} - e^{-j\omega 2} \right) e^{-j\omega 2}$$

$$H(e^{j\omega}) = \left(2.04 + .88(e^{j\omega} + e^{-j\omega}) - (e^{j\omega 2} + e^{-j\omega 2}) \right) e^{-j\omega 2}$$

$$H(e^{j\omega}) = [2.04 + 1.76\cos\omega - 2\cos(2\omega)]e^{-j\omega 2}$$

$$H(e^{j\omega}) = A_e(e^{j\omega})e^{-j\omega 2}, \quad A_e(e^{j\omega}) = A_e(e^{-j\omega})$$

Amplitude and Angle Plot

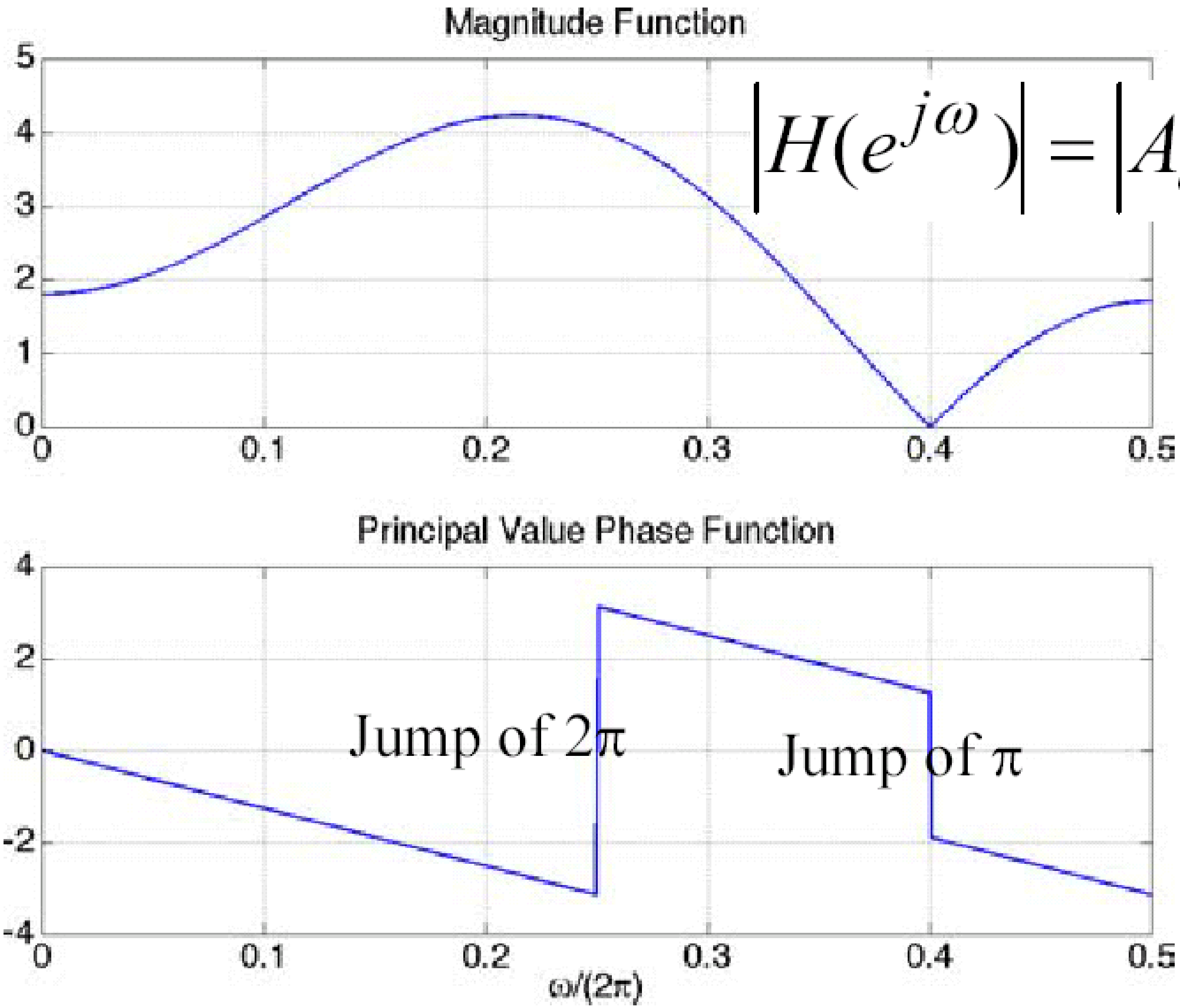


$$A_e(e^{j\omega}) = A_e(e^{-j\omega})$$

$$-\omega 2$$

Magnitude and Phase Plot

$$\left|H(e^{j\omega})\right| = \left|A_e(e^{j\omega})\right|$$



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Pole at *a* , zero at 1/*a**

- Real co-efficient all-pass:

$$A(z) = \prod_k \frac{z^{-1} - a_k^*}{1 - a_k z^{-1}} \prod_k \frac{(z^{-1} - b_k^*)(z^{-1} - b_k)}{(1 - b_k z^{-1})(1 - b_k^* z^{-1})}$$

Generalized Linear Phase

$$\angle\left[H(e^{j\omega})\right] = \underbrace{-\alpha\omega} + \underbrace{\beta}$$

all freq components shifted by the same amount of time

all freq components shifted by the same amount of phase

Generalized linear phase allows

a constant phase shift beside a constant delay.

Generalized Linear Phase

Basic LP system:

$$\underbrace{A(e^{j\omega})}_{\in\mathbb{R}}\, e^{-j(\omega\alpha-\beta)}$$

- constant group delay α (except where $A(e^{j\omega})$ changes sign)

- Common subclasses: $h[2\alpha - n] = h[n]$, $2\alpha = M$

  $h[2\alpha - n] = -h[n]$, $2\alpha = M$